

SUCCESS SECRETS WITH ALEX

Mini-Project: Debug this Code using ChatGPT **Sherin Mary P.J**

PROMPT

You are an experienced JavaScript developer and code reviewer.

Analyze the JavaScript code provided below and perform the following tasks:

1. Identify all syntax, logical, and runtime errors.
2. Explain each issue clearly, including why it causes incorrect behavior.
3. Highlight the exact line(s) where the problem occurs.
4. Provide the corrected version of the code.
5. Ensure the corrected code fulfills the intended requirements:
 - Print all book titles.
 - Collect books that meet the specified rating criteria.
 - Find the longest book based on page count.
 - Categorize books as Long Read, Medium Read, or Quick Read.
 - Display the final results correctly.
6. Follow JavaScript best practices and improve readability where appropriate.
7. Show the expected output after the fixes.

CODE

```
let books = [  
  { title: "Atomic Habits", author: "James Clear", pages: 320, rating: 4.8  
},  
  { title: "Deep Work", author: "Cal Newport", pages: 304, rating: 4.5 },
```

```
    { title: "The 7 Habits of Highly Effective People", author: "Stephen R. Covey", pages: 432, rating: 4.7 },
    { title: "Mindset", author: "Carol Dweck", pages: 320, rating: 4.4 },
    { title: "Can't Hurt Me", author: "David Goggins", pages: 364, rating: 4.6 }
  ];
```

```
// Goal 1: Print all titles
console.log("Top books:");
for (let i = 0; i < books.length; i++) {
  console.log(books[i].title);
}
```

```
// Goal 2: Collect books with rating >= 4.7
let fiveStarGreats = [];
```

```
for (let j = 0; j < books.length; j++) {
  let b = books[j];

  if (b.rating >= 4.7) {
    fiveStarGreats.push(b.title);
  }
}
```

```
// Goal 3: Find the longest book
let longest = null;
```

```
for (let k = 0; k < books.length; k++) {
  let current = books[k];

  if (longest === null || current.pages > longest.pages) {
    longest = current;
  }
}
```

```
// Goal 4: Summary based on page count
console.log("\nSummary:");
```

```
for (let x = 0; x < books.length; x++) {
  let bk = books[x];
```

```

    if (bk.pages >= 400) {
        console.log(bk.title + " = long read.");
    } else if (bk.pages >= 300) {
        console.log(bk.title + " = medium read.");
    } else {
        console.log(bk.title + " = quick read.");
    }
}

// Final Output
console.log("\nFive-star greats:", fiveStarGreats);

if (longest !== null) {
    console.log("Longest book:", longest.title, "-", longest.pages,
"pages");
} else {
    console.log("No data.");
}

```

OUTPUT

Top books:

Atomic Habits

Deep Work

The 7 Habits of Highly Effective People

Mindset

Can't Hurt Me

Summary:

Atomic Habits = medium read.

Deep Work = medium read.

The 7 Habits of Highly Effective People = long read.

Mindset = medium read.

Can't Hurt Me = medium read.

Five-star greats: ['Atomic Habits', 'The 7 Habits of Highly Effective People']

Longest book: The 7 Habits of Highly Effective People - 432 pages